Project 3 - Climate dynamics

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Part 1.

1.a)

Given

$$x' = -\frac{1}{10}x - 3y - 3z$$
$$y' = \frac{24}{5}x - \frac{61}{10}y - \frac{51}{10}z$$
$$y' = -\frac{9}{5}x + 6y + 5z$$

The corresponding matrix A such that **x'** $= A\mathbf{x}$ is

$$A = \begin{bmatrix} -\frac{1}{10} & -3 & -3 \\ \frac{24}{5} & -\frac{61}{10} & -\frac{51}{10} \\ -\frac{9}{5} & 6 & 5 \end{bmatrix}$$

Its eigenvalues are $\lambda_1 = -1$, $\lambda_2 = -\frac{1}{10} - 3i$, $\lambda_3 = -\frac{1}{10} + 3i$ and the corresponding eigenvectors are $v_1 = \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}$, $v_2 = \begin{bmatrix} -\frac{1}{2} \\ -1 - \frac{i}{2} \\ 1 \end{bmatrix}$, $v_3 = \begin{bmatrix} -\frac{1}{2} \\ -1 + \frac{i}{2} \\ 1 \end{bmatrix}$

The formula for the solution is a linear combination of $v_1 e^{\lambda_1 t}$ and of the real and imaginary part of $v_2 e^{\lambda_2 t}$. After the computation we find that

$$\mathbf{f}(t) = ae^{-t}\begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} + be^{-t/10}\begin{bmatrix} -\frac{1}{2}\cos(3t) \\ -\left(\cos(3t) + \frac{1}{2}\sin(3t)\right) \\ \cos(3t) \end{bmatrix} + ce^{-t/10}\begin{bmatrix} \frac{1}{2}\sin(3t) \\ -\left(\frac{1}{2}\cos(3t) - \sin(3t)\right) \\ -\sin(3t) \end{bmatrix}$$

where a, b, c are such that

$$x(0) = x_0$$
$$y(0) = y_0$$
$$z(0) = z_0$$

, which means that

$$a = z_0 + 2x_0$$
$$b = -2x_0$$
$$c = -2z_0 - 2y_0$$

1.b)

We can now compare the obtained formula with the numerical results of the system. For the starting point $(1, 1, 1)$, which means $a = 3$, $b = -2$, $c = 4$. The plots are in Fig.1, 2, 3

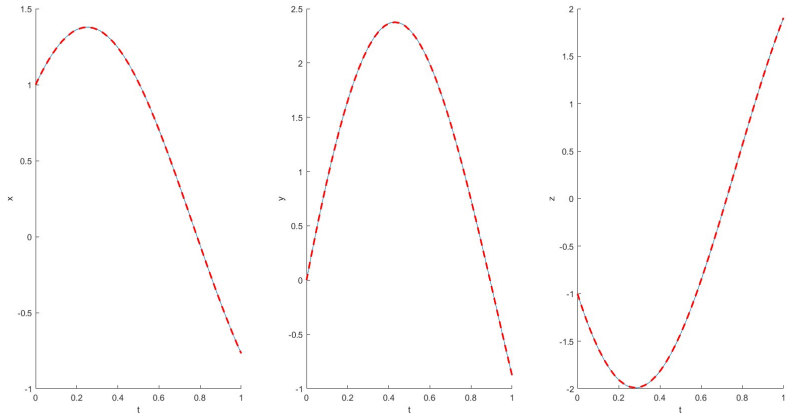


Figure 1: Plot of the system evolution in the time-variable 2D plane from the starting point (1,0,-1). The thin continuous blue line represents the numerical solution, the wide dotted red line represents the formula we found for the solution

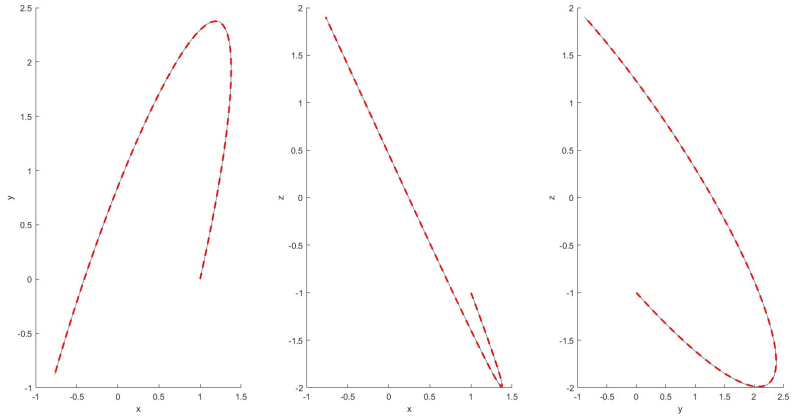


Figure 2: Plot of the system evolution in the variable-variable 2D plane from the starting point (1,0,-1). The thin continuous blue line represents the numerical solution, the wide dotted red line represents the formula we found for the solution

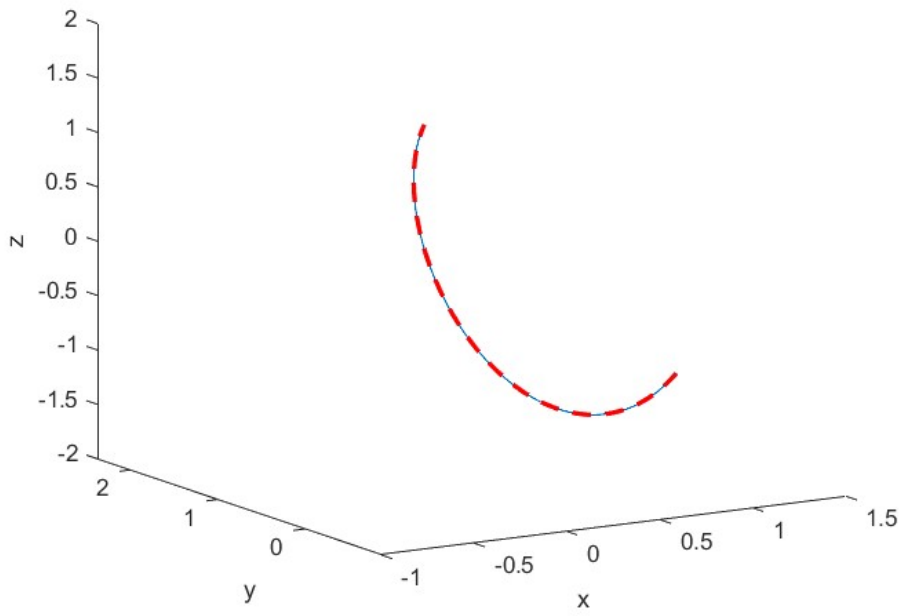


Figure 3: Plot of the system evolution in the 3D plane from the starting point (1,0,-1). The thin continuous blue line represents the numerical solution, the wide dotted red line represents the formula we found for the solution

For the starting point $(1, 1, 1)$, which means $a = 1$, $b = -2$, $c = 2$. The plots are in Fig.4, 5, 6

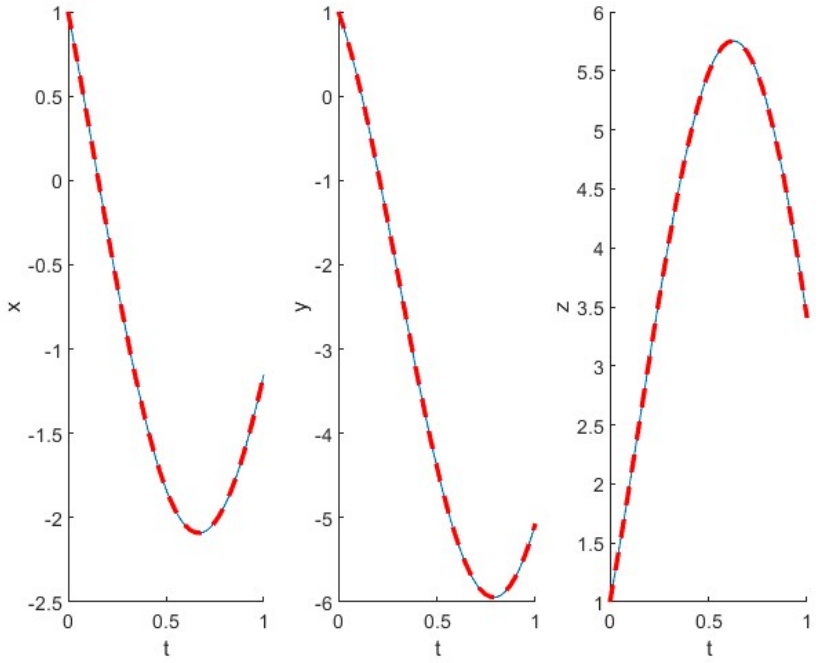


Figure 4: Plot of the system evolution in the time-variable 2D plane from the starting point (1,1,1). The thin continuous blue line represents the numerical solution, the wide dotted red line represents the formula we found for the solution

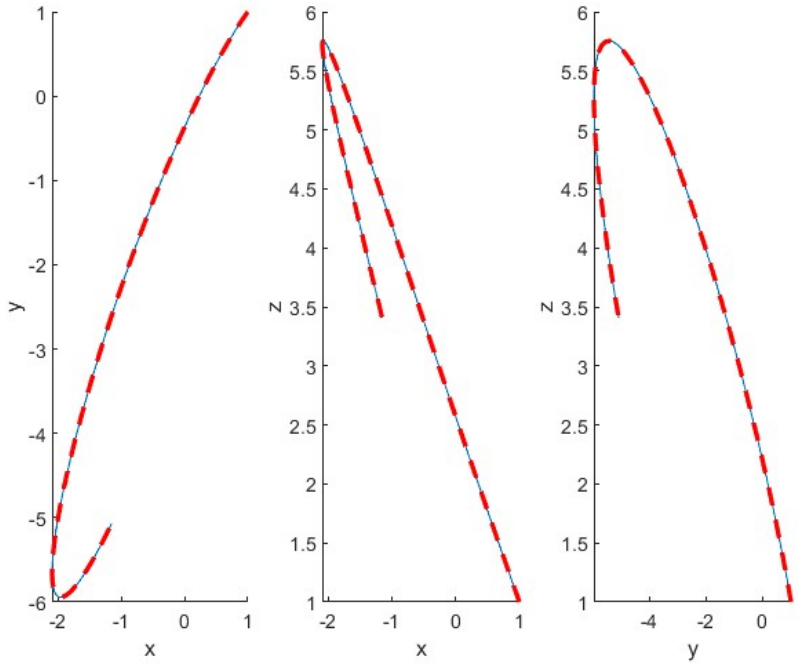


Figure 5: Plot of the system evolution in the variable-variable 2D plane from the starting point (1,1,1). The thin continuous blue line represents the numerical solution, the wide dotted red line represents the formula we found for the solution

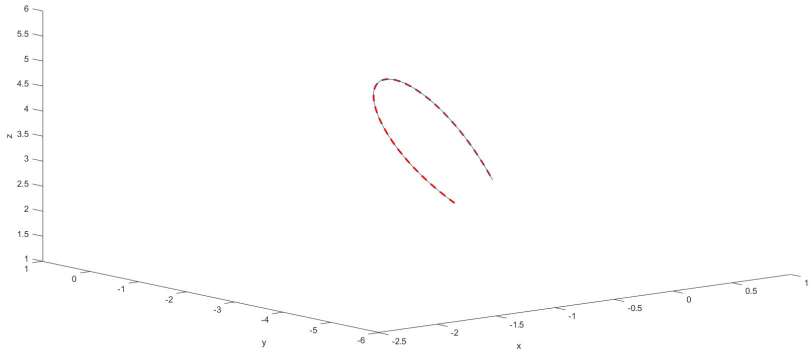


Figure 6: Plot of the system evolution in the 3D plane from the starting point (1,1,1). The thin continuous blue line represents the numerical solution, the wide dotted red line represents the formula we found for the solution

As we can see from both starting points the formula perfectly represents the dynamic of the system.

2.a)

We have the system

$$
\begin{aligned}
u' &= 2(a-6) + (a-12)u - 15v - 3w - 3u^2 - 12uv - 3u^2v \\
v' &= 4a + 5 + (2a+1)u + (a-6)v + (a-1)w - u^2 + 4uw + u^2w \\
w' &= -4a - 2(a+1)u + 9v - w + 3w^2 - 6vw + 3vw^2
\end{aligned}
$$

To find a fix point solution to the system that is independent of the parameter a we need $(\bar{u}, \bar{v}, \bar{w})$ such that it is a fix point $\forall a$. If such a point exists it has to solve

$$\begin{cases} 2a + \bar{u}a = 0 \\ 4a + 2a\bar{u} + a\bar{v} + a\bar{w} = 0 \\ -4a - 2a\bar{u} = 0 \end{cases}$$

$$\begin{cases} \bar{u} = -2 \\ \bar{v} = -\bar{w} \end{cases}$$

We now put these in the original system with $\bar{u}', \bar{v}', \bar{w}' = 0$ and find that the fixed point independent of the parameter is $(-2, -1, 1)$.

2.b)

Given that this fix point is independent of a, his branch in the a-u plane has to be a horizontal line at $u = -2$. It is shown in Fig.7

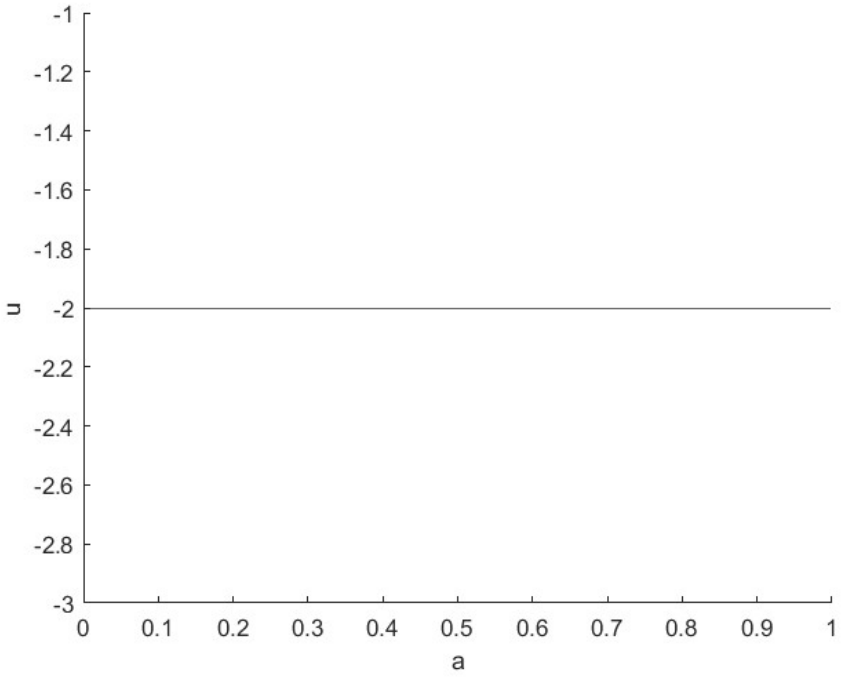


Figure 7: branch on the a-u plane given by the fix point independent of a

2.c)

Now we create a program based on the method described in the section 4.3 of the lectures to find the branches of the bifurcation diagram. The program takes in input the system, a fix point to compute the branch from and the time span of the ode solver. It then creates the ode representing the branches of the system as shown in section 4.3 and numerically solves it, giving in output the resulting branch. The computation of the branch for the point $(-2, -1, 1)$ is shown in Fig.8. We can see that the numerical construction accurately recreates the branch.

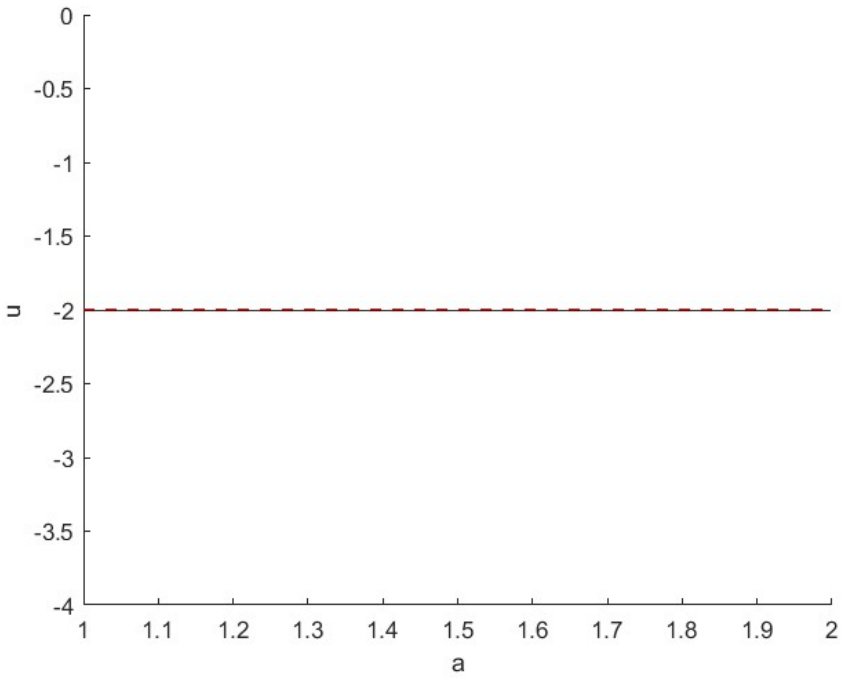


Figure 8: branch on the a-u plane given by the fix point independent of a. In black there is the theoretical line as in Fig.7, the red dotted line is the numerical construction of the branch

2.d)

The complete bifurcation diagram is in Fig.9

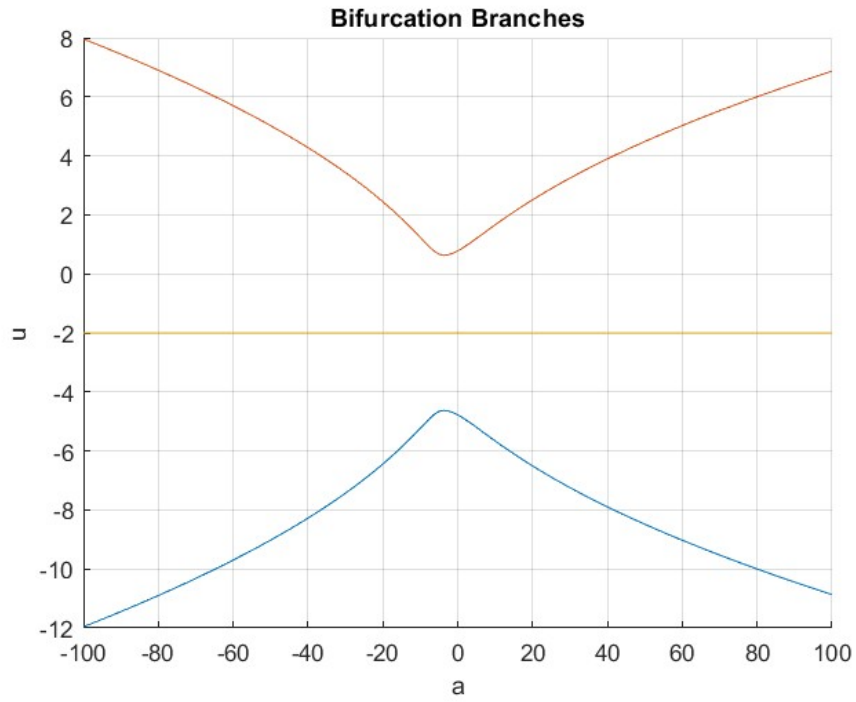


Figure 9: Complete bifurcation diagram of the system

2.e)

We can investigate the stability of the branches by looking at the eigenvalues of the Jacobian matrix. The graph representing the stability of the fixed points is Fig.10

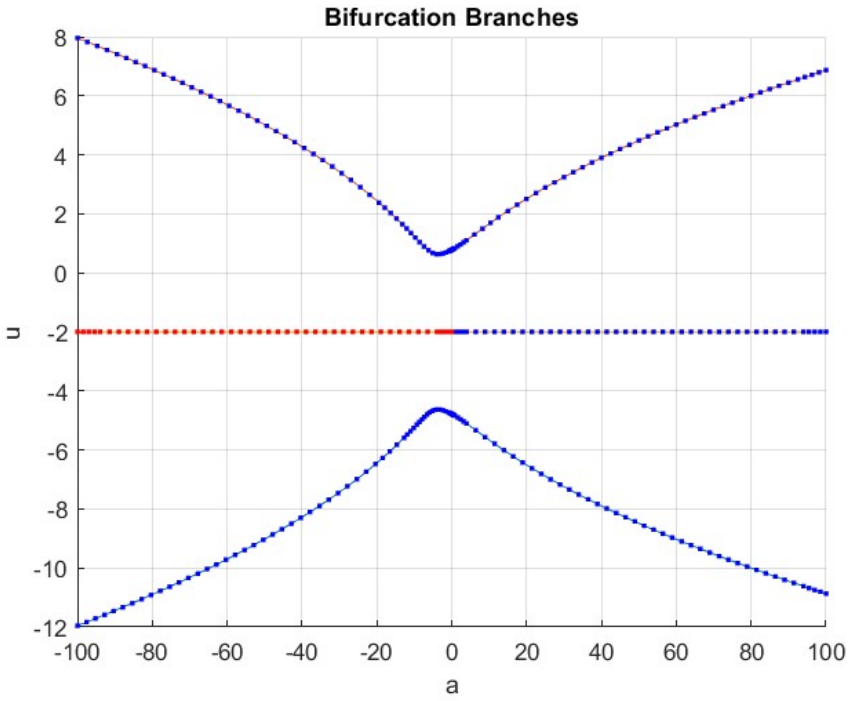


Figure 10: Complete bifurcation diagram of the system with stability properties. In red the stable points, in blue the unstable points.

We can clearly notice a bifurcation point in $a = 0$. The eigenvalues of the Jacobian evaluated in $(a, u, v, w) = (-0.0001, -2, -1, 1)$ are

$$\begin{bmatrix} -0.0001 + 3.0000i \\ -0.0001 - 3.0000i \\ -1.0000 + 0.0000i \end{bmatrix}$$

, in $(a, u, v, w) = (0, -2, -1, 1)$ they are

$$\begin{bmatrix} -0.0000 + 3.0000i \\ -0.0000 - 3.0000i \\ -1.0000 + 0.0000i \end{bmatrix}$$

and in $(a, u, v, w) = (0.0001, -2, -1, 1)$ they are

$$\begin{bmatrix} 0.0001 + 3.0000i \\ 0.0001 - 3.0000i \\ -1.0000 + 0.0000i \end{bmatrix}$$

.

As we can see in $a = 0$ a pair of complex conjugates have real part crossing from negative to positive. This means that in $a = 0$ there is a Hopf bifurcation.

2.f)

We can verify the found stability properties of the branches using direct numerical solution of the system. For a choice of a negative a the resulting plot is Fig.11. As we can see from all starting points close to fixed points the system converges to $(-2, -1, 1)$ or diverges completely. This confirm the stability properties found for these branches in the previous point.

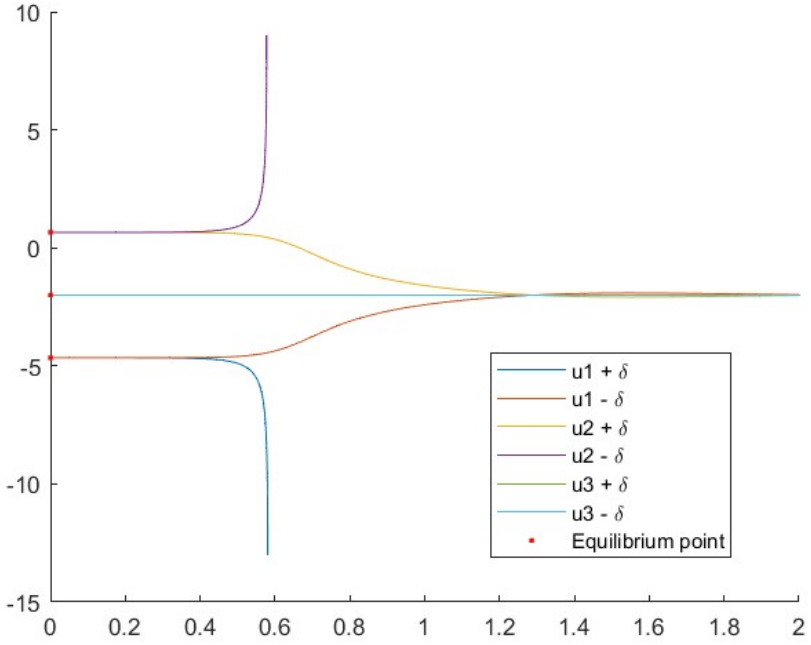


Figure 11: dynamic of the system on the t-u plane with a negative value for a.

For a choice of a positive a the resulting plot is Fig.12. As we can see from all starting points close to fixed points the system diverges. This confirm the stability properties found for these branches in the previous point.

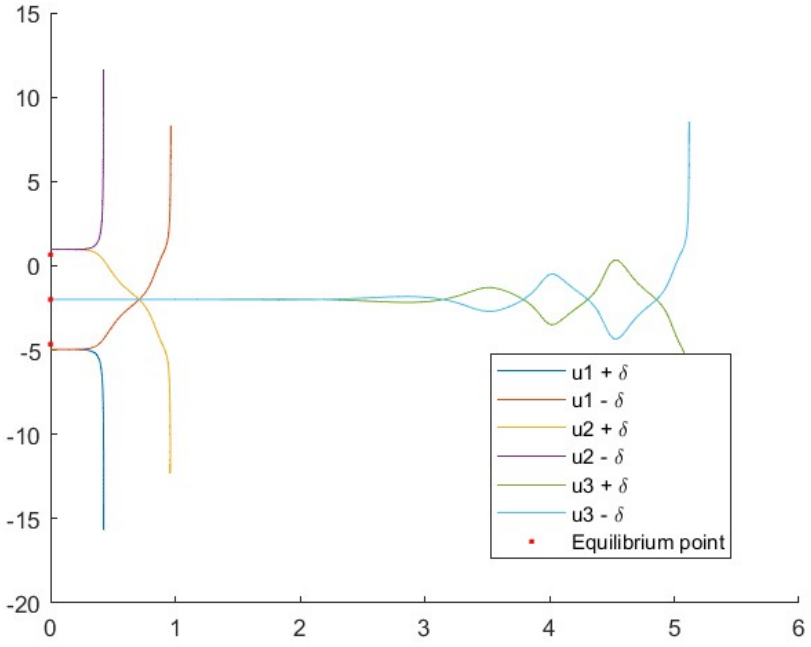


Figure 12: dynamic of the system on the t-u plane with a positive value for a.

Part 2.

1)

The dimensional system is

$$\dot{T}_1 = \begin{cases} q\,(T_2 - T_1) + \lambda\,(T_p - T_1)\,, & q \geq 0 \\ |q|\,(T_3 - T_1) + \lambda\,(T_p - T_1)\,, & q < 0 \end{cases}$$
$$\dot{T}_2 = \begin{cases} \frac{q}{V}\,(T_3 - T_2) + \lambda\,(T_e - T_2)\,, & q \geq 0 \\ \frac{|q|}{V}\,(T_1 - T_2) + \lambda\,(T_e - T_2)\,, & q < 0 \end{cases}$$
$$\dot{T}_3 = \begin{cases} q\,(T_1 - T_3) + \lambda\,(T_p - T_3)\,, & q \geq 0 \\ |q|\,(T_2 - T_3) + \lambda\,(T_p - T_3)\,, & q < 0 \end{cases}$$
$$\dot{S}_1 = \begin{cases} q\,(S_2 - S_1) - \Phi_N, & q \geq 0 \\ |q|\,(S_3 - S_1) - \Phi_N, & q < 0 \end{cases}$$
$$\dot{S}_2 = \begin{cases} \frac{q}{V}\,(S_3 - S_2) + \frac{\Phi_N + \Phi_S}{V}, & q \geq 0 \\ \frac{|q|}{V}\,(S_1 - S_2) + \frac{\Phi_N + \Phi_S}{V}, & q < 0 \end{cases}$$
$$\dot{S}_3 = \begin{cases} q\,(S_1 - S_3) - \Phi_S, & q \geq 0 \\ |q|\,(S_2 - S_3) - \Phi_S, & q < 0, \end{cases}$$

where

$$q = k\,[\alpha\,(T_3 - T_1) - \beta\,(S_3 - S_1)]\,.$$

We define $T_N \equiv T_2 - T_1$, $T_S \equiv T_2 - T_3$, $S_N \equiv S_2 - S_1$, $S_S \equiv T_2 - T_3$, $T_0 \equiv T_e - T_p$ and the model becomes

$$\dot{T}_N = \dot{T}_2 - \dot{T}_1 = \begin{cases} -\frac{q}{V}T_S + \lambda T_0 - q\,T_N - \lambda T_N, & q \geq 0 \\ -\frac{|q|}{V}T_N + \lambda T_0 - |q|(T_N - T_S) - \lambda T_N, & q < 0 \end{cases}$$
$$\dot{T}_S = \dot{T}_2 - \dot{T}_3 = \begin{cases} -\frac{q}{V}T_S + \lambda T_0 - q(T_S - T_N) - \lambda T_S, & q \geq 0 \\ -\frac{|q|}{V}T_N + \lambda T_0 - |q|\,T_S - \lambda T_S, & q < 0 \end{cases}$$
$$\dot{S}_N = \dot{S}_2 - \dot{S}_1 = \begin{cases} -\frac{q}{V}S_S + \frac{\Phi_N + \Phi_S}{V} - qS_N + \Phi_N, & q \geq 0 \\ -\frac{|q|}{V}S_N + \frac{\Phi_N + \Phi_S}{V} - |q|(S_N - S_S) + \Phi_N, & q < 0 \end{cases}$$
$$\dot{S}_S = \dot{S}_2 - \dot{S}_3 = \begin{cases} -\frac{q}{V}S_S + \frac{\Phi_N + \Phi_S}{V} - q(S_S - S_N) + \Phi_S, & q \geq 0 \\ -\frac{|q|}{V}S_N + \frac{\Phi_N + \Phi_S}{V} - |q|S_S + \Phi_S, & q < 0 \end{cases}$$

where

$$q = k\,[\alpha\,(T_N - T_S) - \beta\,(S_N - S_S)]\,.$$

We can now scale the model using $\vartheta = \frac{k\alpha}{V\lambda}T$, $s = \frac{k\beta}{V\lambda}S$, $v = V$, $\tau = t\lambda$ and

$$\frac{q}{V\lambda} = \frac{k\,[\alpha\,(T_N - T_S) - \beta\,(S_N - S_S)]}{V\lambda} = \vartheta_N - \vartheta_S + s_S - s_N = \xi$$

$$\varphi_1 = \tfrac{\beta k(v+1)}{\lambda^2 v^2}\Phi_N + \tfrac{\beta k}{\lambda^2 v^2}\Phi_S,$$
$$\varphi_2 = \tfrac{\beta k}{\lambda^2 v^2}\Phi_N + \tfrac{\beta k(v+1)}{\lambda^2 v^2}\Phi_S.$$

$$\tfrac{d\vartheta_N}{d\tau} = \begin{cases} -\frac{q}{V\lambda}(\vartheta_S + v\,\vartheta_N) + \vartheta_0 - \vartheta_N = -\xi(\vartheta_S + v\,\vartheta_N) + \vartheta_0 - \vartheta_N, & \xi \geq 0 \\ \frac{|q|}{V\lambda}(-\vartheta_N(1+v) + v\,\vartheta_S) + \vartheta_0 - \vartheta_N = |\xi|(-\vartheta_N(1+v) + v\,\vartheta_S) + \vartheta_0 - \vartheta_N, & \xi < 0 \end{cases}$$
$$\tfrac{d\vartheta_S}{d\tau} = \begin{cases} \frac{q}{V\lambda}(-\vartheta_S(1+v) + v\,\vartheta_N) + \vartheta_0 - \vartheta_S = \xi(-\vartheta_S(1+v) + v\,\vartheta_N) + \vartheta_0 - \vartheta_S, & \xi \geq 0 \\ -\frac{|q|}{V\lambda}(\vartheta_N + v\,\vartheta_S) + \vartheta_0 - \vartheta_S = -|\xi|(\vartheta_N + v\,\vartheta_S) + \vartheta_0 - \vartheta_S, & \xi < 0 \end{cases}$$
$$\tfrac{ds_N}{d\tau} = \begin{cases} -\frac{q}{v\lambda}(s_S - v\,s_N) + \frac{k\beta}{v\lambda^2}\left(\frac{\Phi_N + \Phi_S}{v} + \Phi_N\right) = -\xi(s_S - v\,s_N) + \varphi_1, & \xi \geq 0 \\ \frac{|q|}{v\lambda}(s_S v - (1+v)s_N) + \frac{k\beta}{v\lambda^2}\left(\frac{\Phi_N + \Phi_S}{v} + \Phi_N\right) = |\xi|(s_S v - (1+v)s_N) + \varphi_1, & \xi < 0 \end{cases}$$
$$\tfrac{ds_S}{d\tau} = \begin{cases} \frac{q}{v\lambda}(s_N v - (1+v)s_S) + \frac{k\beta}{v\lambda^2}\left(\frac{\Phi_N + \Phi_S}{v} + \Phi_S\right) = \xi(s_N v - (1+v)s_S) + \varphi_2, & \xi \geq 0 \\ -\frac{|q|}{v\lambda}(s_S v + s_N) + \frac{k\beta}{v\lambda^2}\left(\frac{\Phi_N + \Phi_S}{v} + \Phi_S\right) = -|\xi|(s_S v + s_N) + \varphi_2, & \xi < 0 \end{cases}$$

2.a)

i)

We can solve the dimensional system numerically using as parameters

$\alpha = 1.5 \times 10^{-4}$ K^{-1}
$\beta = 0.8 \times 10^{-3}$psu^{-1}
$T_p = 0°$C
$T_e = 30°$C
$\lambda = 12.9 \times 10^{-10}$ s^{-1}
$k = 1.5 \times 10^{-6}$ s^{-1}
$V = 2\Phi_S = 0.9 \times 10^{-10}$ psu s$^{-1}\Phi_N = 1.35 \times 10^{-10}$ psu s^{-1}

and a time span of [0 s, 100000000000 s] or [0 yrs, , 3171 yrs]. We can see the resulting dynamics in Fig.13

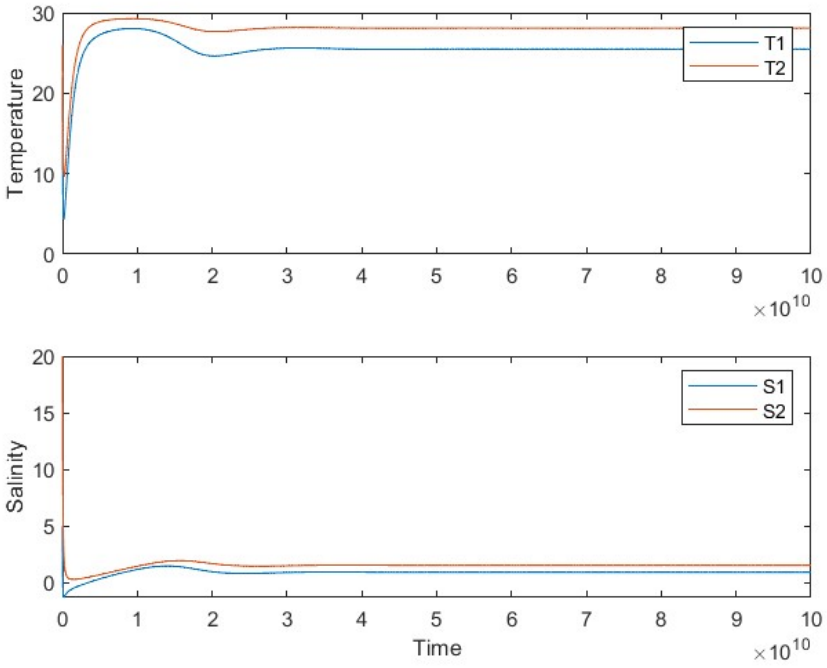


Figure 13: evolution of the three-box ocean dimensional system.

ii)

We can repeat the computation on the same system with scaled units, using the transformations described in 1). The resulting plot in in Fig.14

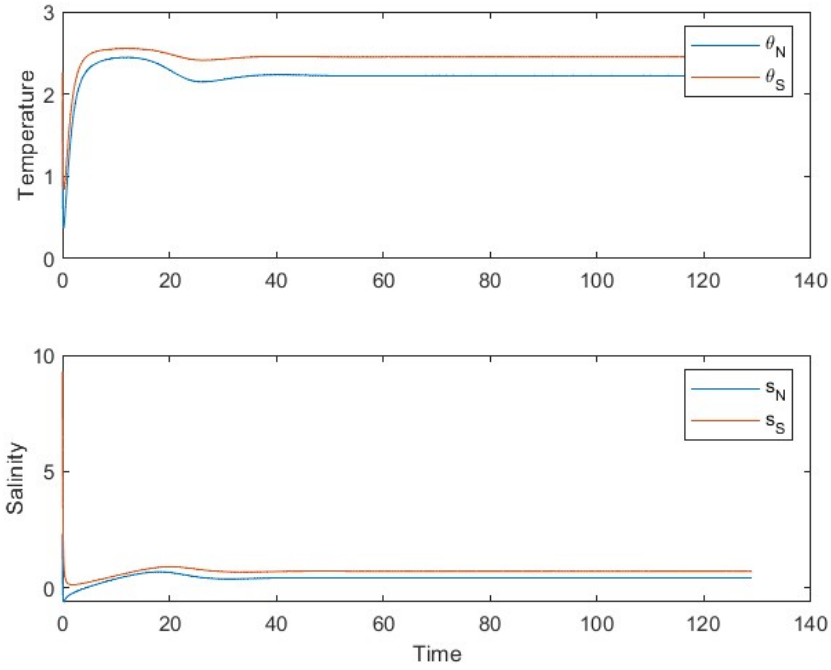


Figure 14: evolution of the three-box ocean non-dimensional system.

iii)

We can now verify the correctness of our computations by plotting the numerical solutions from part i) and ii) together on the scaled time interval, using the scaling units to convert the unscaled numerical solution into a scaled one. The result is in Fig.15

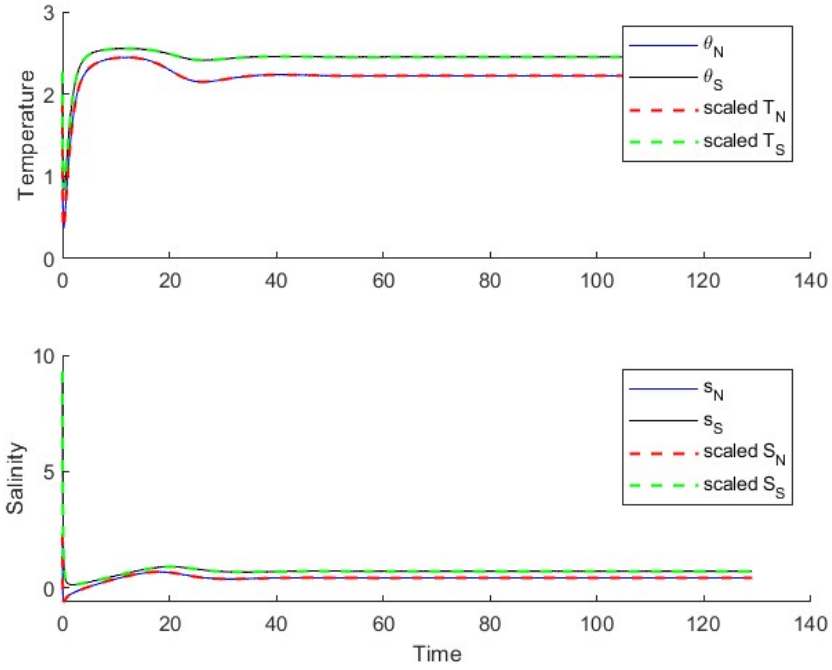


Figure 15: Comparison of the unscaled and scaled numerical solutions on the scaled units.

As we can see the two dynamics overlap perfectly.

2.b)

Now we create a program based on the method described in the section 4.3 of the lectures to find the branches of the bifurcation diagram. The program takes in input the system, a fix point to compute the branch from and the time span of the ode solver. It then creates the ode representing the branches of the system as shown in section 4.3 and numerically solves it, giving in output the resulting branch.

We now test the code on the system

$$\begin{aligned} x' &= \mu + xw, \\ y' &= y - x, \\ z' &= z - y, \\ w' &= w - z. \end{aligned}$$

We can find the exact solution for the branches.

$$\begin{cases} \mu + xw &= 0, \\ y - x &= 0, \\ z - y &= 0, \\ w - z &= 0. \end{cases} \Rightarrow \begin{cases} \mu &= -x^2, \\ y &= x, \\ z &= y, \\ w &= z. \end{cases} \Rightarrow \begin{cases} x &= \pm\sqrt{-\mu}, \\ y &= \pm\sqrt{-\mu}, \\ z &= \pm\sqrt{-\mu}, \\ w &= \pm\sqrt{-\mu}. \end{cases}$$

We only have fixed points for $\mu < 0$.

We can now plot the result of our program compared to the exact solution in Fig.16

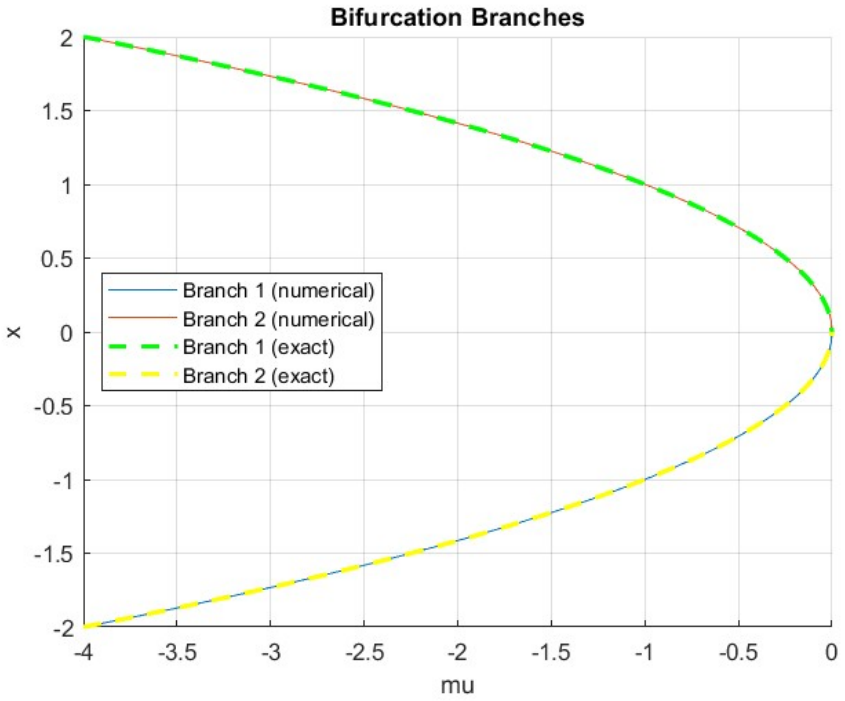


Figure 16: Comparison of the bifurcation diagram computed using the exact solution and the method from 4.3 in the lectures on the μ-x plane.

We can see that they precisely overlap each other.

2.c)

We can see the branches corresponding to the Northern sinking branch and Southern sinking branch using the program in Fig.17

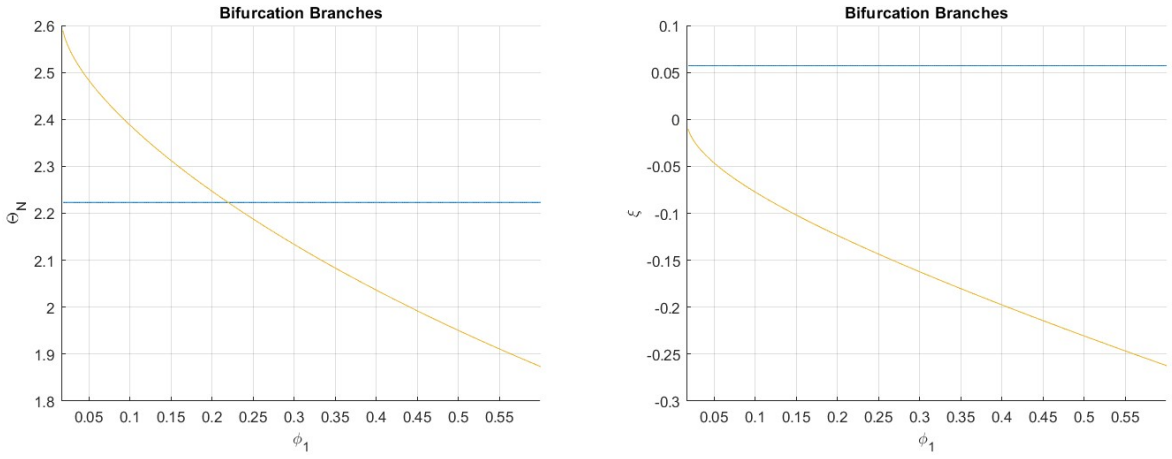


Figure 17: Bifurcation diagrams for the three-box ocean model.

2.d)

We can investigate the stability of the branches by looking at the eigenvalues of the Jacobian matrix in the fixed points. The results of the stability analysis are in Fig.18

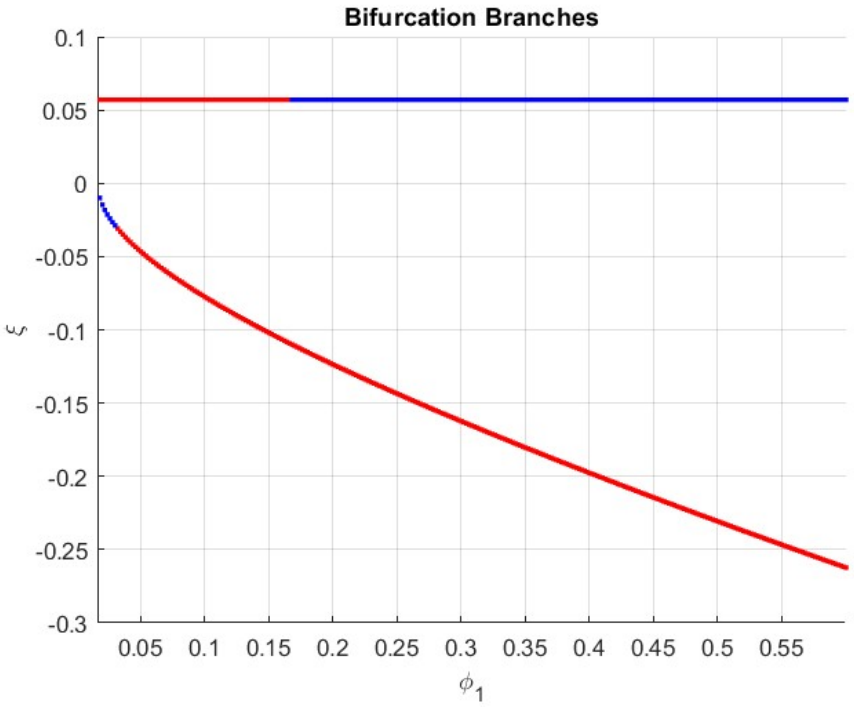


Figure 18: Bifurcation diagrams for the three-box ocean model, in red there are the stable fixed points and in blue the unstable ones.

Around where the stability switches for the Northern sinking branch the eigenvalues of the points are

$$\begin{bmatrix} -4.3657 + 0.0000i \\ -0.9937 + 0.0000i \\ 0.0018 + 0.2653i \\ 0.0018 - 0.2653i \end{bmatrix} \quad \begin{bmatrix} -4.3987 + 0.0000i \\ -0.9933 + 0.0000i \\ -0.0052 + 0.2643i \\ -0.0052 - 0.2643i \end{bmatrix}$$

Doing the same for the Southern sinking branch gives

$$\begin{bmatrix} -5.0910 + 0.0000i \\ -0.9958 + 0.0000i \\ -0.0076 + 0.1345i \\ -0.0076 - 0.1345i \end{bmatrix} \quad \begin{bmatrix} -5.0481 + 0.0000i \\ -0.9965 + 0.0000i \\ 0.0087 + 0.1256i \\ 0.0087 - 0.1256i \end{bmatrix}$$

We can see that for both cases there is a pair of complex conjugates with real part going from negative to positive and vice-versa. This means that those points are Hopf bifurcation points.

2.e)

We can validate the found stability properties of the branches by direct numerical solution of the system. The starting points can be seen in Fig.19 and their corresponding dynamics are 20

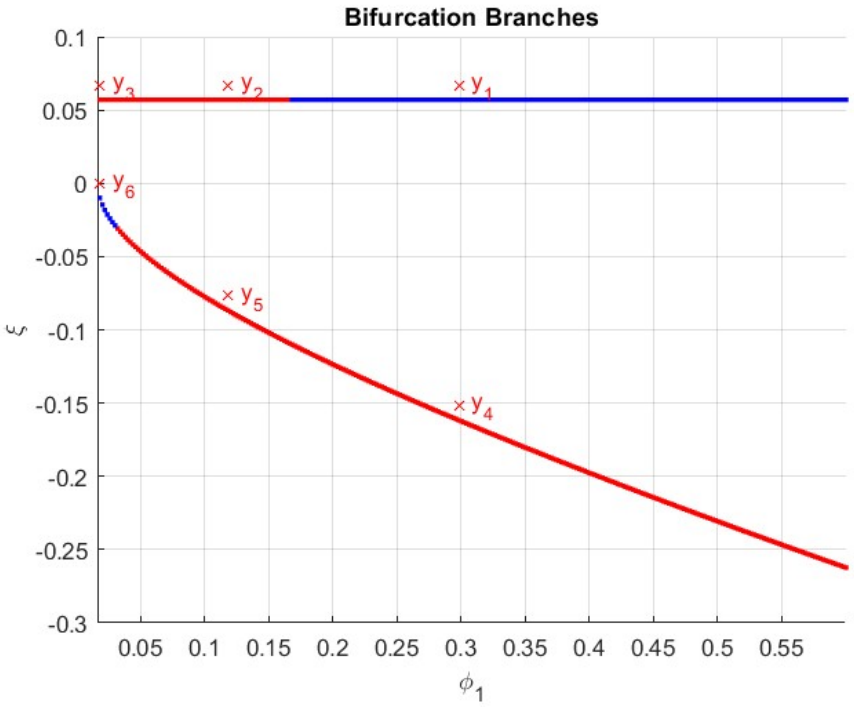


Figure 19: Starting points for the stability validation.

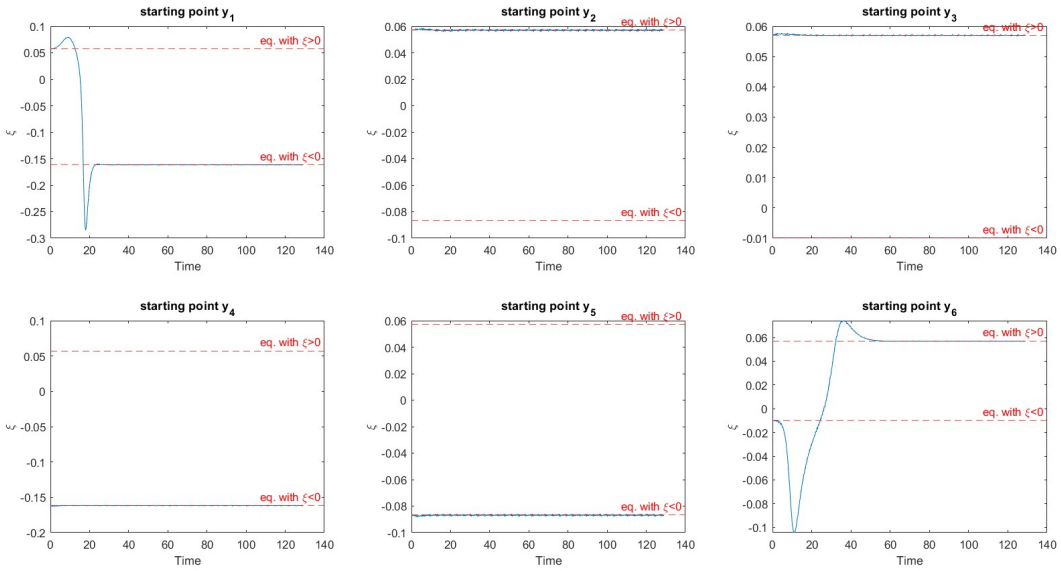


Figure 20: Dynamics of the system for a selection of starting points.

As we expected looking at Fig.19 from the starting points y_1 and y_6 converge to the fixed point further away from them, while the others converge to the closest fixed point.

2.f)

We can find the numerical values for μ_1 and μ_2 using Newton's method. From a point close to the Hopf bifurcation point we use as equations the ones from the scaled system using the equilibrium condition and adding to them the fact that $\det(\mathbf{J} - \omega i I)$ must have real and imaginary part $= 0$, with $\mathbf{J}$ Jacobian of the system and ω a real number.

The resulting Hopf bifurcation points are in $\mu_2 = 0.1670$ and $\mu_1 = 0.0310$.

We can convert these values into numerical values for the dimensional parameter ϕ_N: $\phi_{N,2} = 2.7871e-10$, $\phi_{N,1} = 2.7384e-11$. Comparing them to the ones indicated in 21 shows that our calculations are correct.

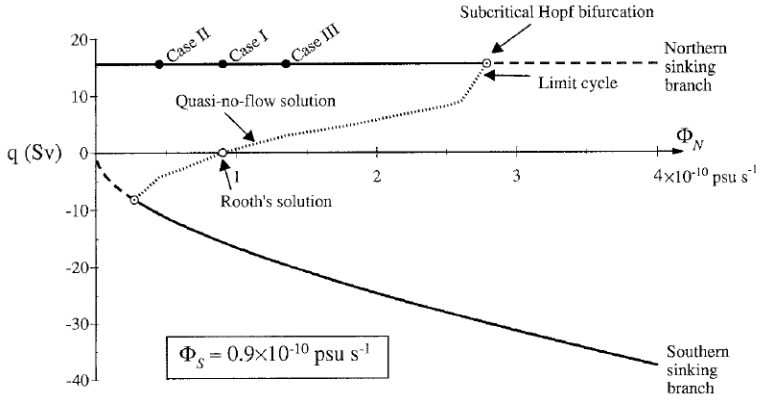


Figure 21: bifurcation diagram in figure 5 in Scott's paper

2.g)

Starting from the same point as the bifurcation point for the Northern sinking branch but with the bifurcation parameter ϕ_N slightly increased ($\phi_N = \phi_{N,2} + 0.015 \times 10^{-10}$ psu s^{-1}) the system moves as shown in Fig.22

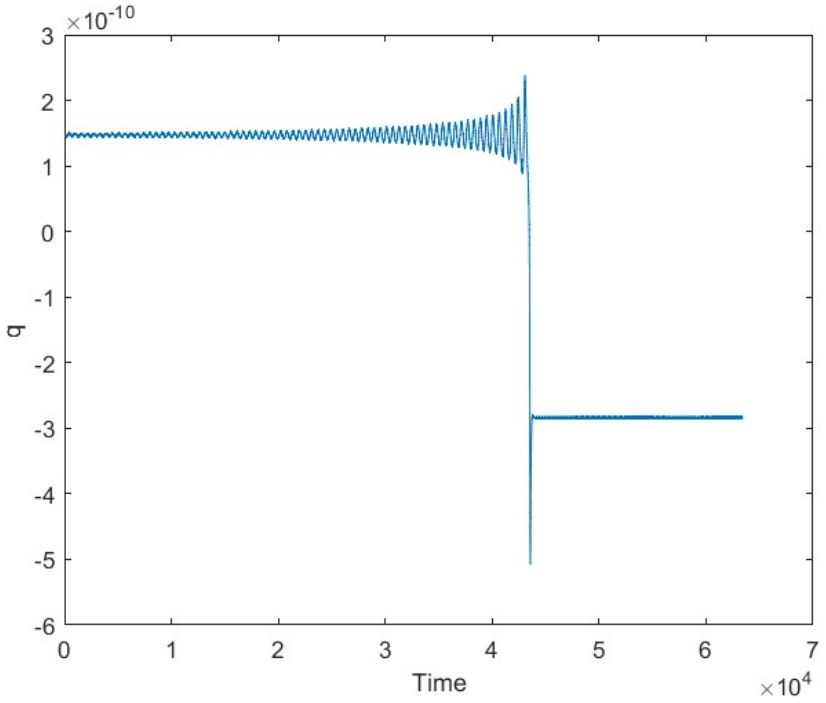


Figure 22: System dynamic for the switch from the Northern sinking branch to the Southern sinking branch

Starting from the same point as the bifurcation point for the Southern sinking branch but with the bifurcation parameter ϕ_N slightly increased ($\phi_N = \phi_{N,1} + 0.01 \times 10^{-11}$ psu s^{-1}) the system moves as shown in Fig.23

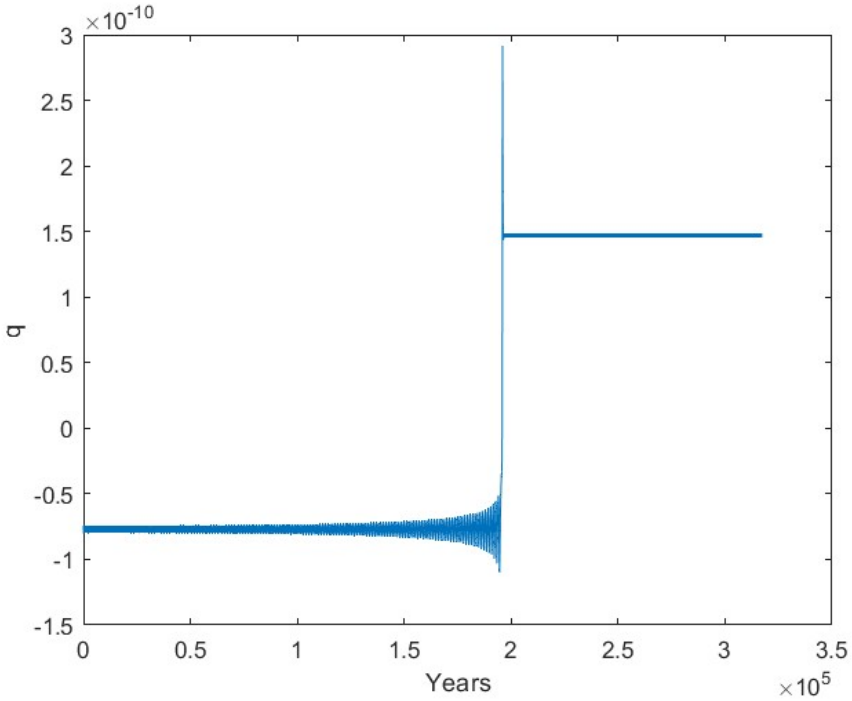


Figure 23: System dynamic for the switch from the Southern sinking branch to the Northern sinking branch

The switch from the Northern sinking branch to the Southern sinking branch takes approximately 44 000 years, the one from the Southern sinking branch to the Northern sinking branch takes approximately 200 000 years. This result is consistent with the result indicated in figure 6 of Scott's paper and may vary depending on the choice for the increment on the bifurcation parameter.